

HANSHIN LEE

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EDUCATION

M.S in Data Science

September 2025 - Present

Pusan National University

Overall GPA: 4.00/4.00

B.S in Physics (Minor in Economics)

March 2019 - August 2025

Pusan National University

Overall GPA: 3.49/4.00 (Major GPA: 3.51/4.00)

RELEVANT COURSEWORK

Letter grades are shown in parentheses. Courses currently underway are marked “”*

Graduate:

- **Math & Statistics:** Linear Algebra I (A), Mathematical Analysis I (A), Mathematical Statistics I (A), Data Science Math (A), Linear Algebra II (*), Mathematical Analysis II (*), Mathematical Statistics II (*)
- **Data Science:** Introduction to Data Science (A), Machine Learning (A)
- **Computer Science:** Data Computing (A), Data Programming (A), Data Visualization (*)

Undergraduate:

- **Math & Statistics:** Calculus I (A), Calculus II (A), Mathematical Physics I (B+), Econometrics (A), Mathematics for Economics (A), Statistics for Economics (A)
- **Physics:** Mechanics I (A), Mechanics II (A), Electricity and Magnetism I (A), Electricity and Magnetism II (A), Statistical Physics (A), Computational Physics (A)

RESEARCH EXPERIENCE

Graduate Researcher

September 2025 - Present

Advisor: *Prof.* Minwoo Kim (Statistics)

PUBLICATION

“Condition assignment determines the appropriate differential gene expression test in single-cell RNA sequencing studies” Under Review

Hanshin Lee[†], Hanbin Lee[†], Buhm Han^{*}, Minwoo Kim^{*}

Bioinformatics (Oxford University Press)

- Contribute to methodological research on differential gene expression in multi-subject single-cell RNA-seq, distinguishing subject-level from cell-level comparison-label assignment as different sampling designs.
- Re-derived and revised M-estimation arguments, subject- and cell-level variance decompositions, and asymptotic rates $O(1/n)$ versus $O(1/N)$ using Taylor expansion and the central limit theorem.
- Clarified how the valid sampling unit changes the calibration of type I and type II errors for pseudobulk and cell-wise differential-expression methods under two label-assignment scenarios.

SCHOLARSHIPS & FELLOWSHIPS

National Science & Engineering Graduate Stipend

Ministry of Science and ICT (MSIT), Republic of Korea

September 2025 - Present

New Coming Graduate Student Fellowship

Pusan National University, Graduate School of Data Science

September 2025 - August 2026

Academic Scholarship

Pusan National University, Graduate School of Data Science

March 2026 - February 2027

Academic Scholarship

Pusan National University, Department of Physics

September 2019 - August 2020

PERSONAL SKILLS

Programming: Python, R, LaTeX

Languages: Korean (native), English (IELTS 7.0)

ADDITIONAL EXPERIENCE

Military Service

Sergeant, Republic of Korea Navy

July 2020 – March 2022